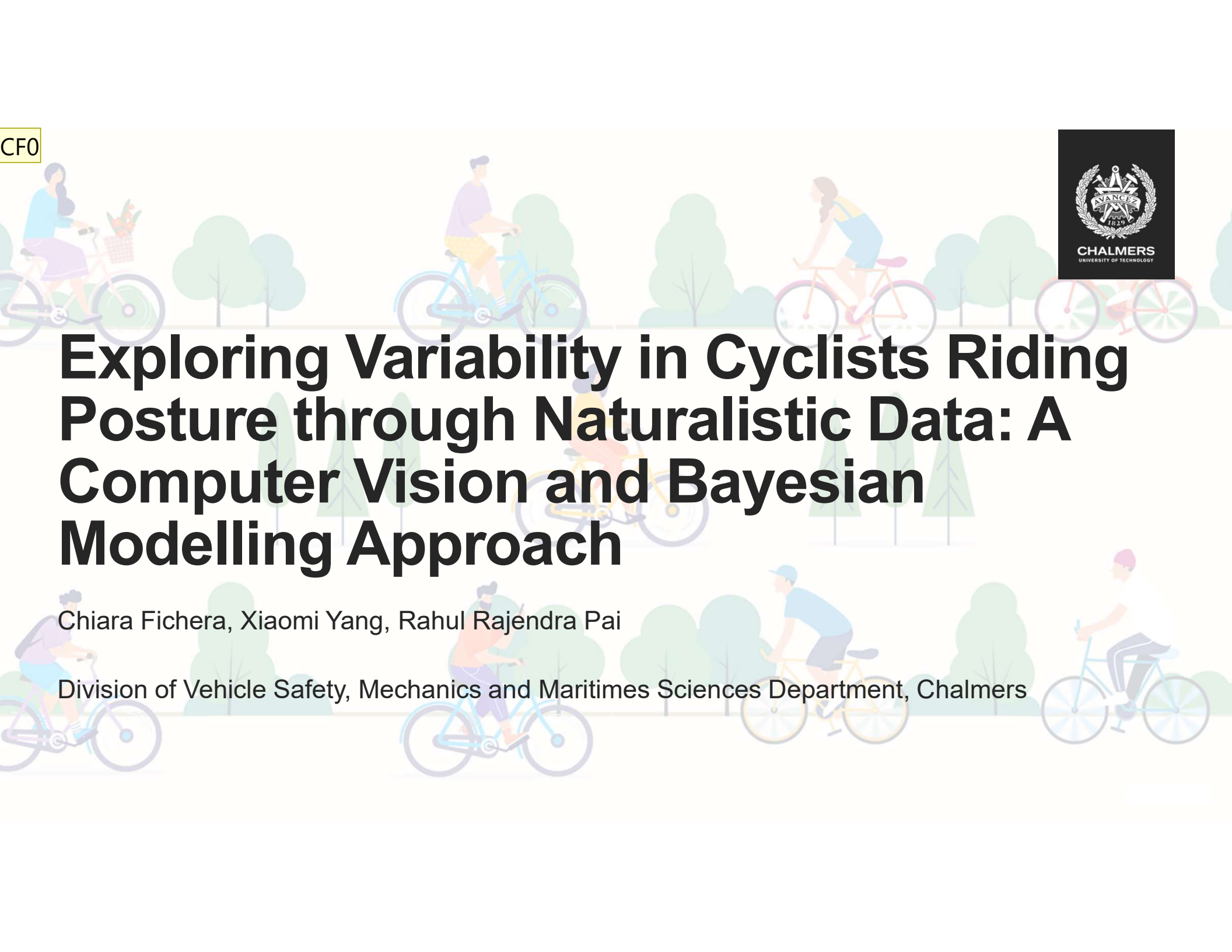




Exploring Variability in Cyclists Riding Posture through Naturalistic Data: A Computer Vision and Bayesian Modelling Approach

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Division of Vehicle Safety, Mechanics and Maritimes Sciences Department, Chalmers

Slide 1

CF0

Very nice image, btw! 😊

Chiara Rosanna Fichera; 2024-05-29T15:08:43.618

RP0 0

Thats my crontribution to the project!

Rahul Rajendra Pai; 2024-05-30T06:56:20.952

Introduction



The severity and type of injuries sustained by cyclists can be influenced by various factors such as riding posture, helmet usage, bike type, and speed.

By understanding how people interact with the vehicle, we can understand initial boundary conditions during a crash and are able to design prevention systems.

Introduction

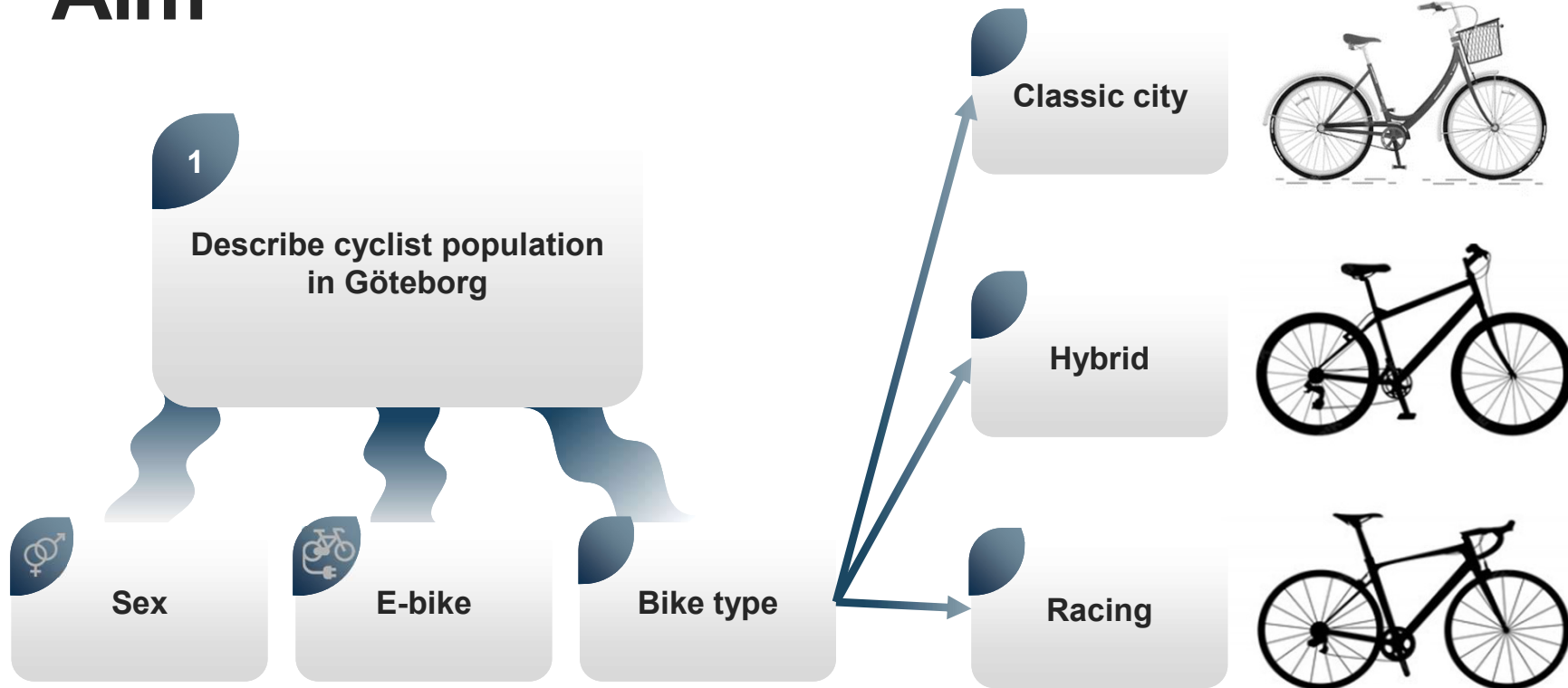


The severity and type of injuries sustained by cyclists can be influenced by various factors such as **riding posture**, **helmet usage**, **bike type**, and speed.

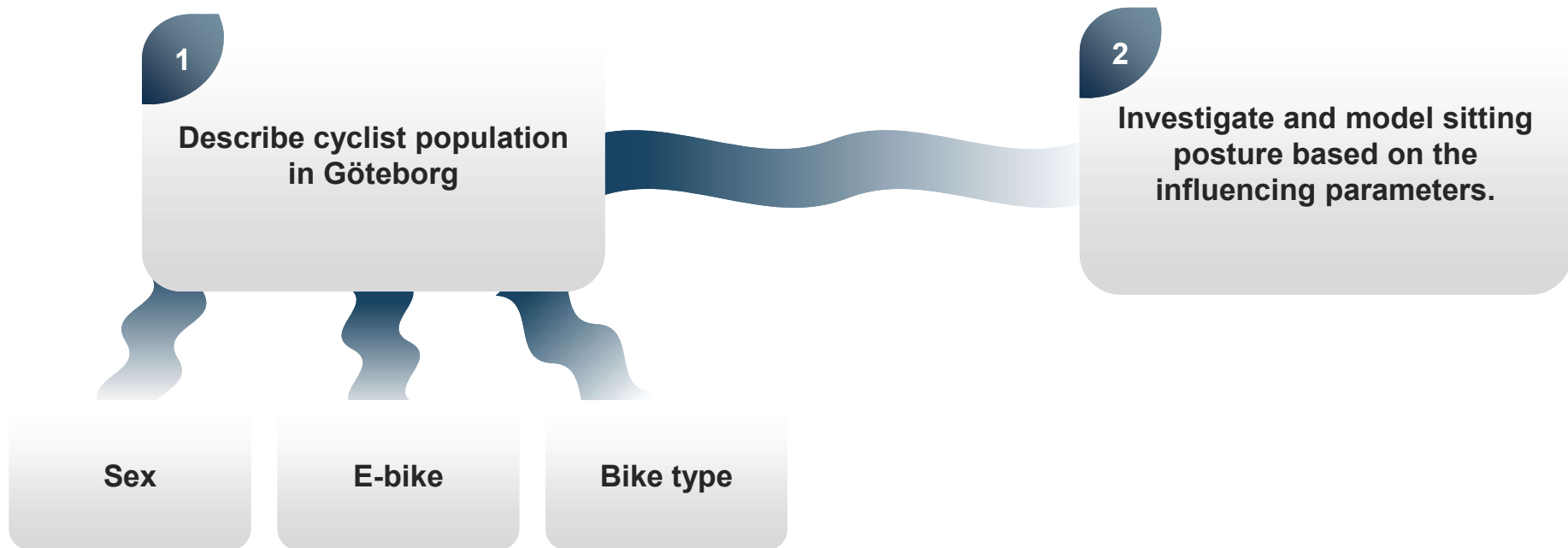
By understanding how people interact with the vehicle, we can understand initial boundary conditions during a crash and are able to design prevention systems.



Aim



Aim



Methods



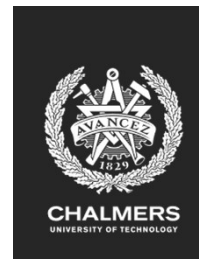
Collect and extract data by video tracking algorithms and annotations.

Descriptive statistics to analyze variability of cyclists

Develop Bayesian regression to analyze and model cyclist posture

Predict sitting position of cyclist based on the developed model

Data Collection



XY0

Slide 7

XY0

Is it bicycle or e-scooter

Xiaomi Yang; 2024-06-03T13:30:14.363

RP0 0

both

Rahul Rajendra Pai; 2024-06-03T13:33:50.169

XY0 1

I'm working on the result figures now! Will make it look better!

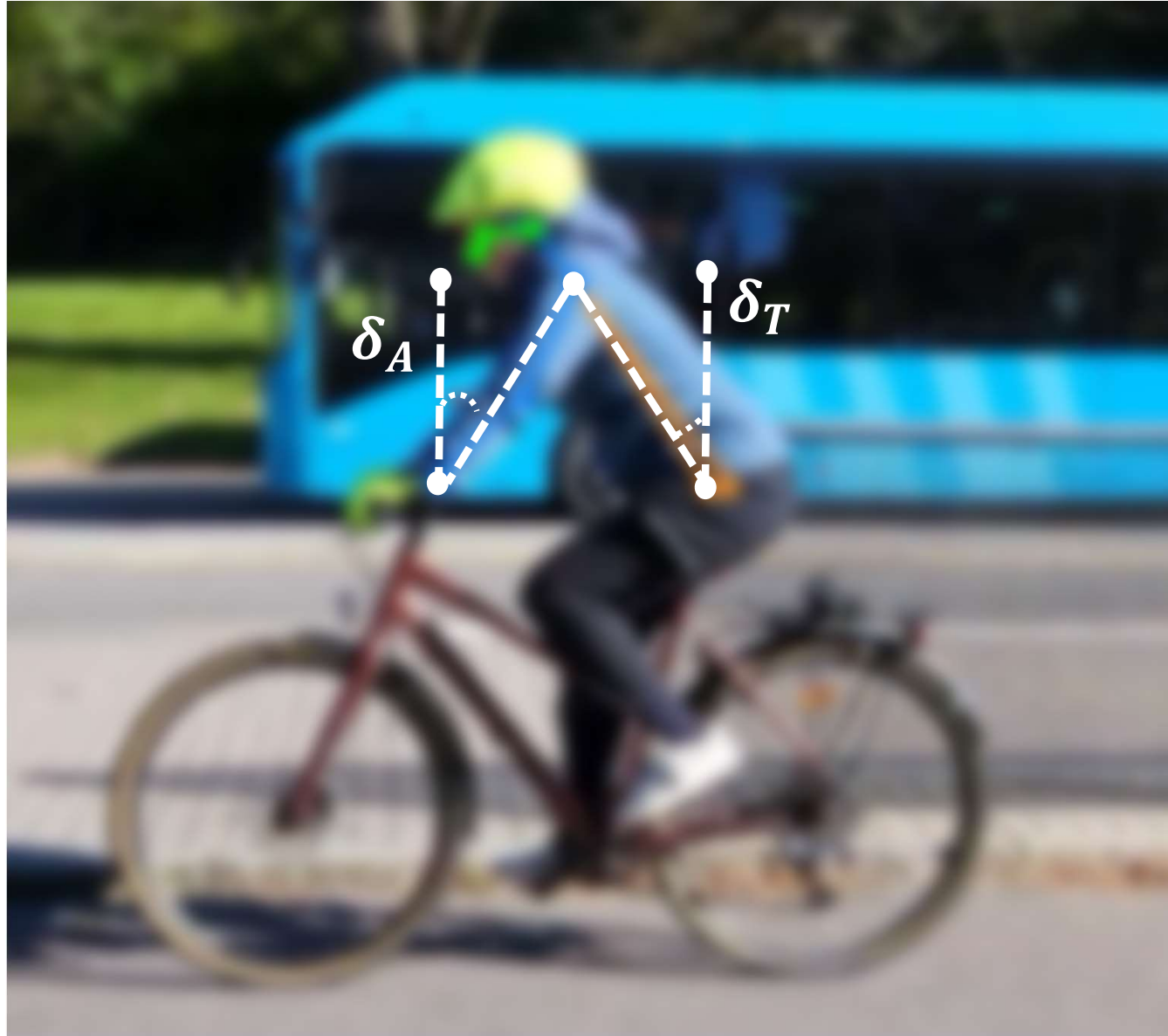
Xiaomi Yang; 2024-06-03T13:34:59.200

CF1

Here maybe add the process of the workflow

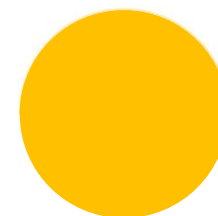
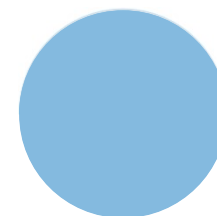
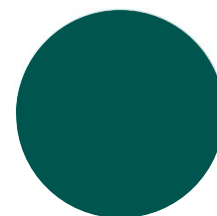
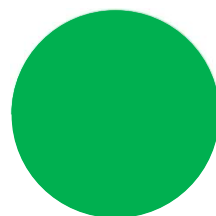
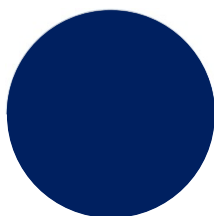
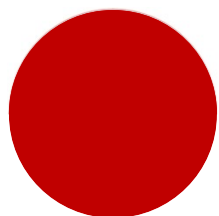
Chiara Rosanna Fichera; 2024-10-03T09:02:14.763

**Compute
Torso
&
Arm angles**



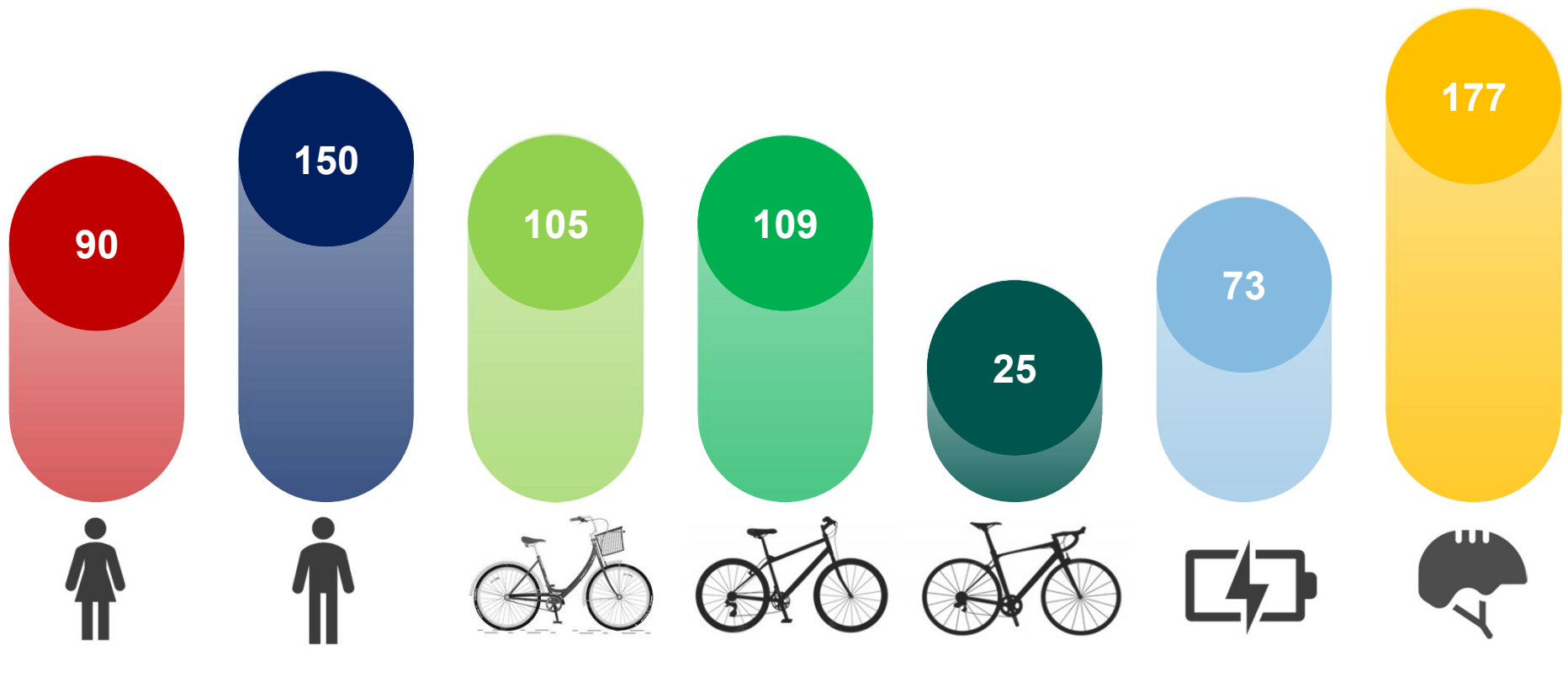
Results

Total sample: 240 Cyclists

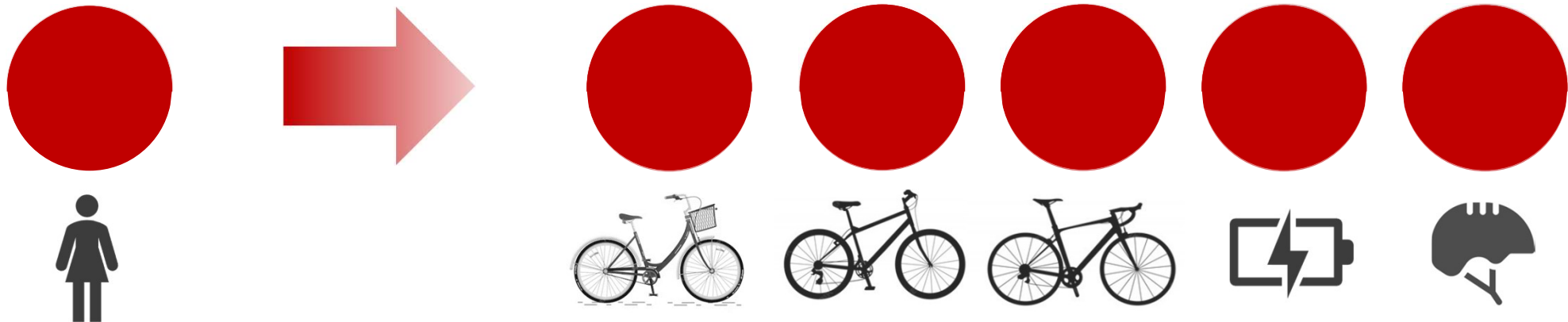


Results

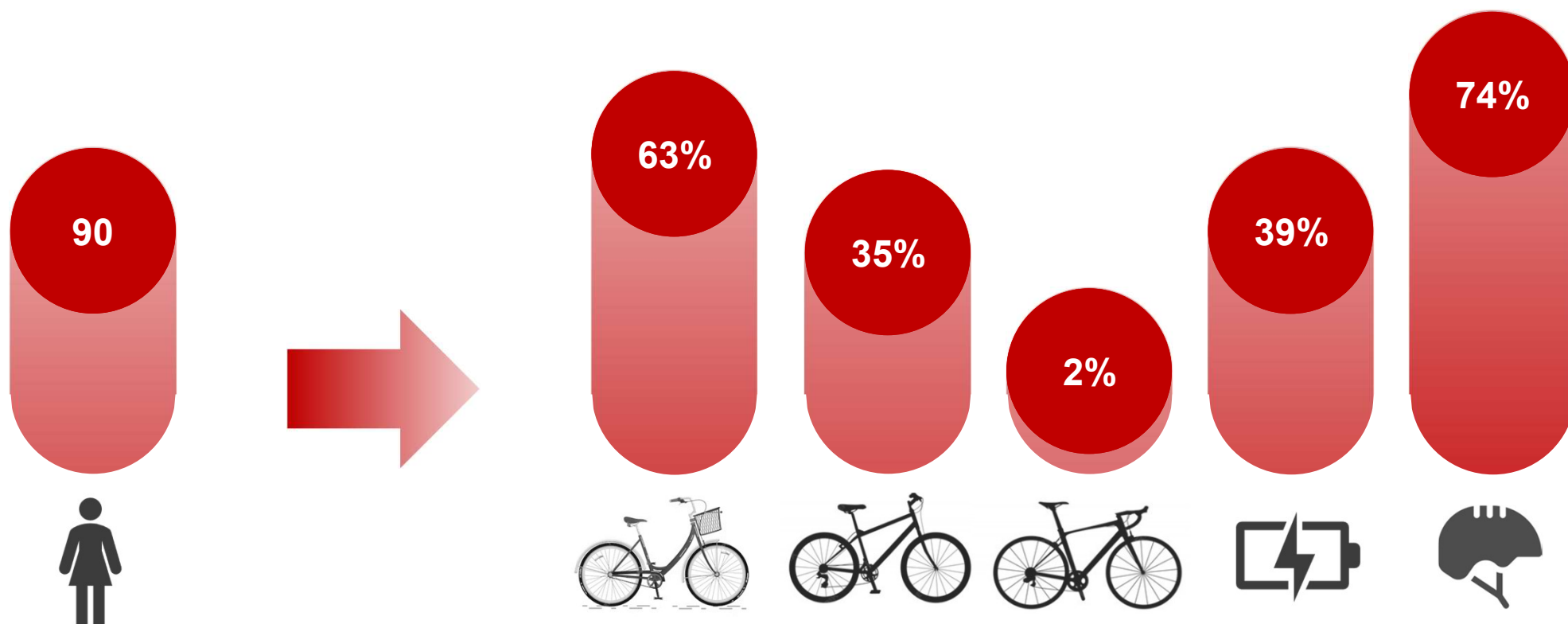
Total sample: 240 Cyclists



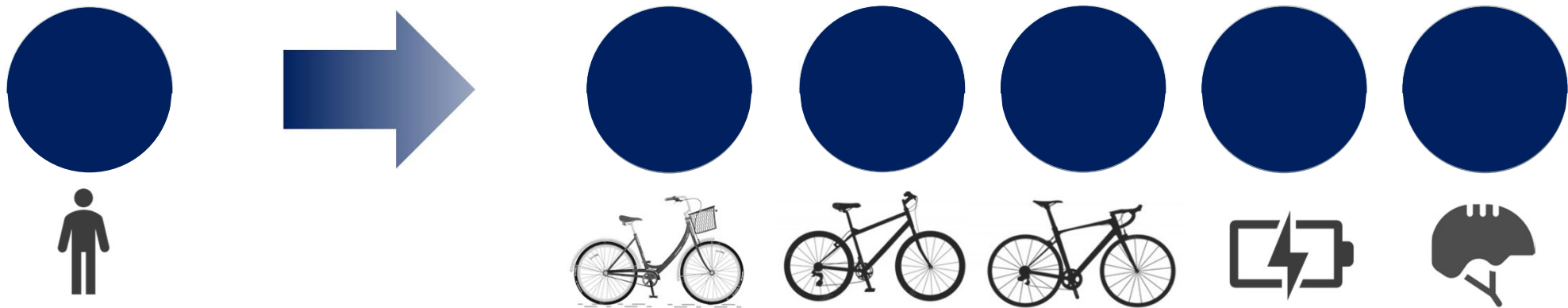
Females vs. Bike type, E-bike, Helmet



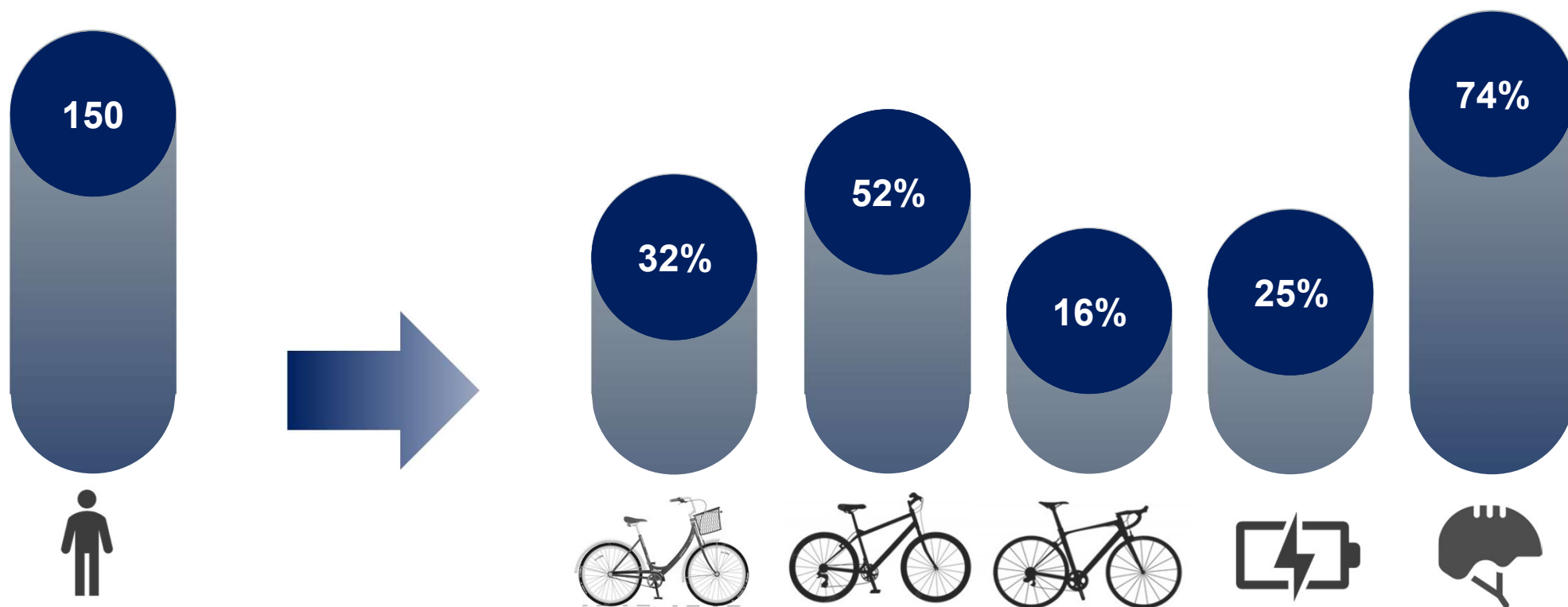
Females vs. Bike type, E-bike, Helmet



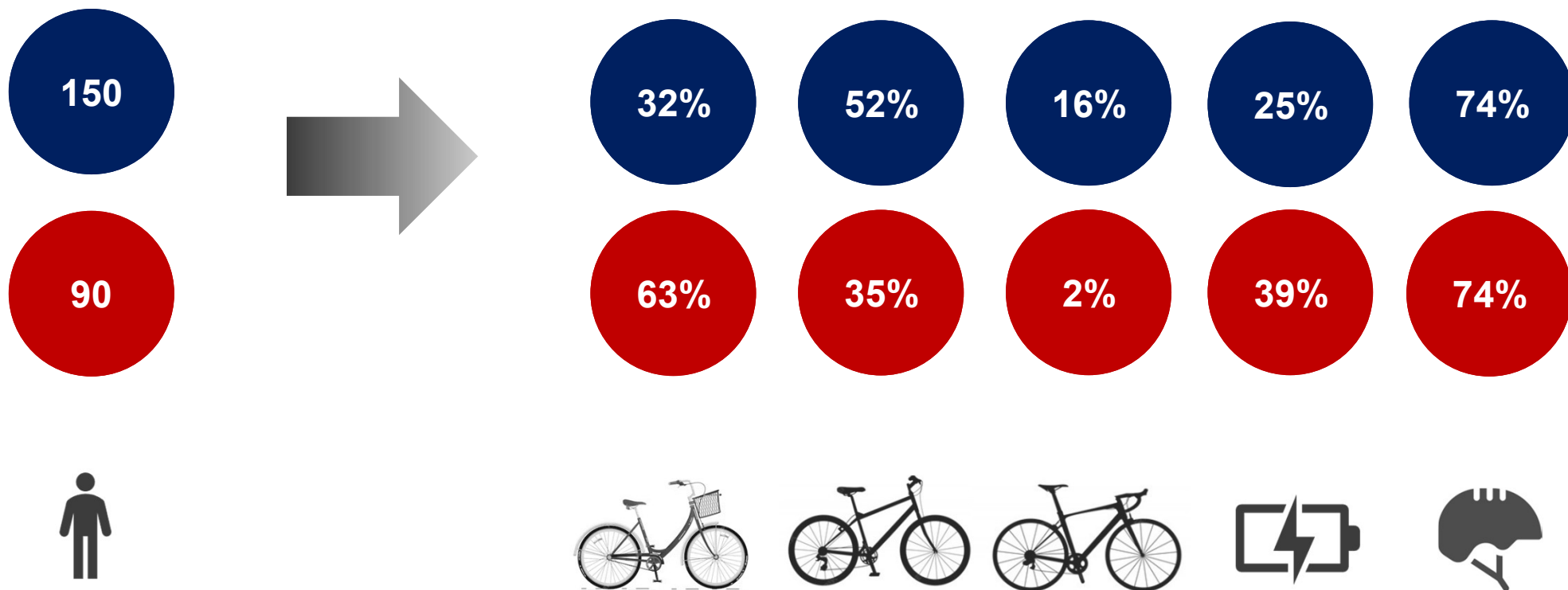
Males vs. Bike type, E-bike, Helmet



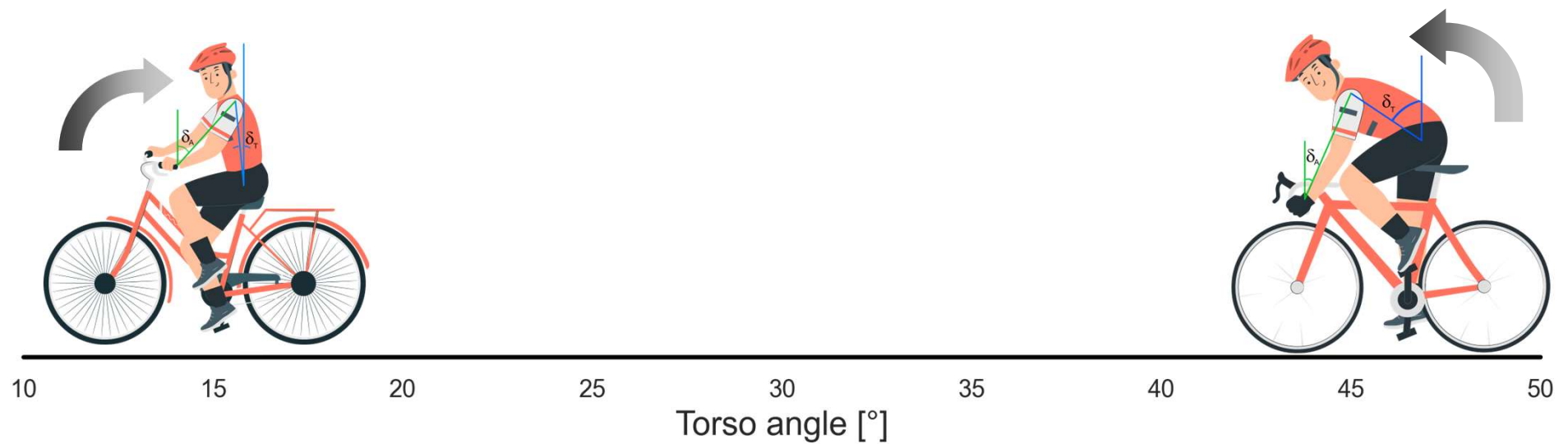
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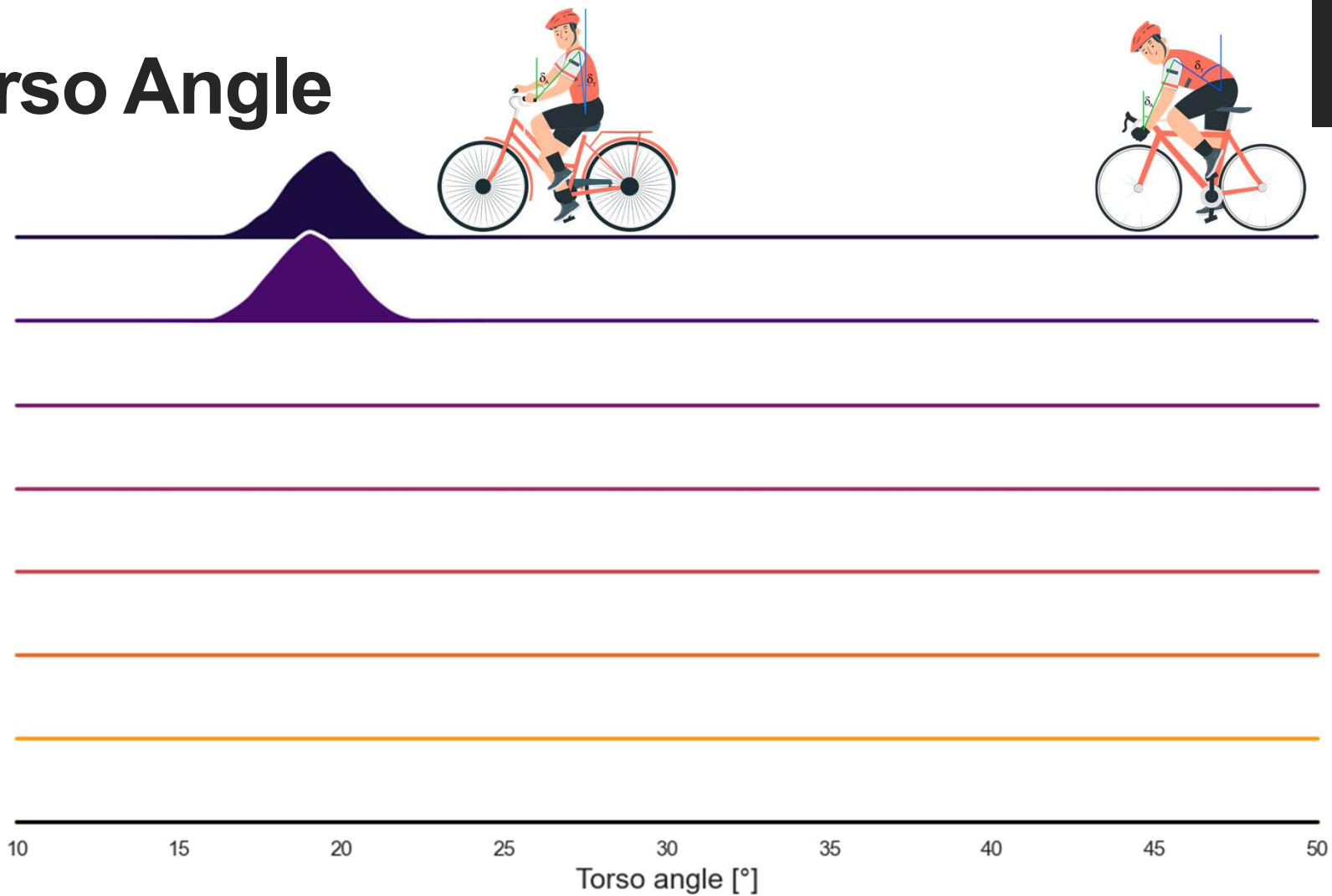
Comparison



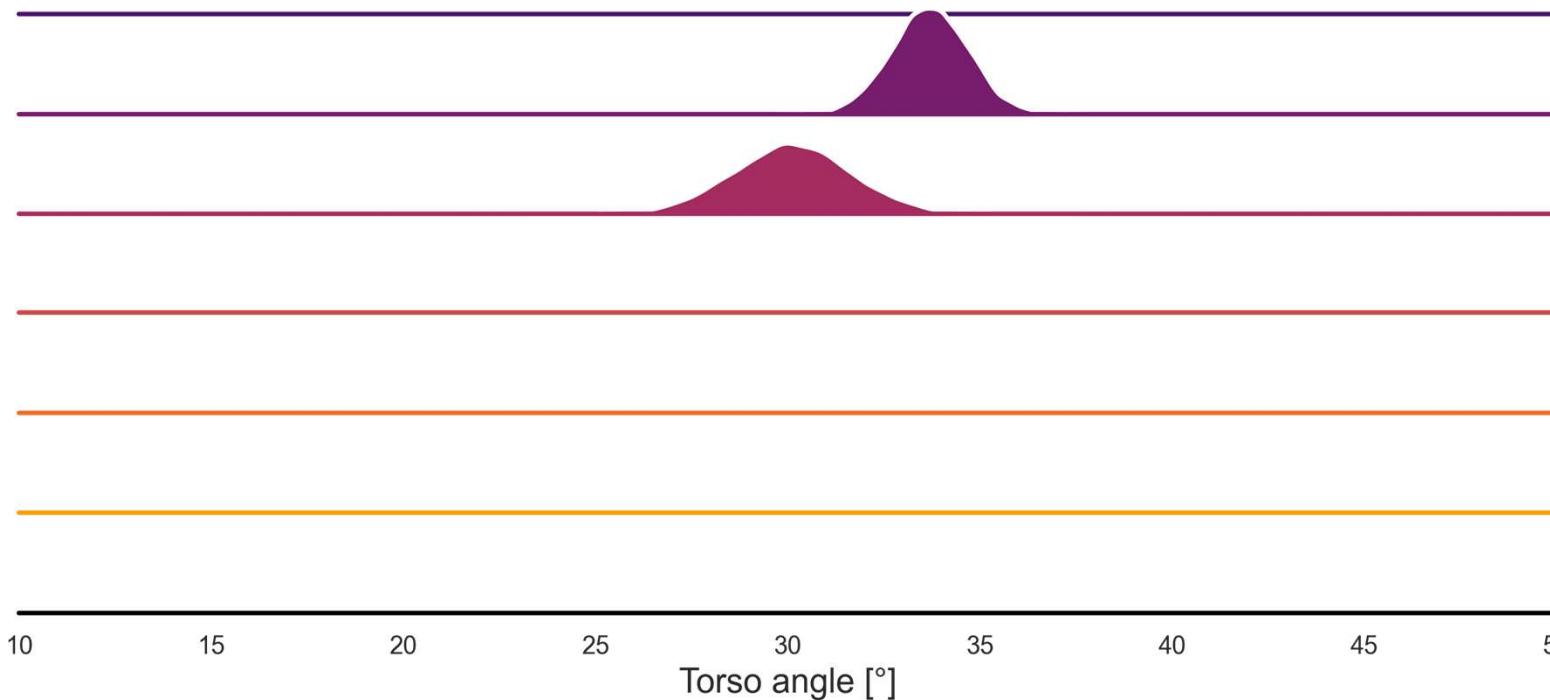
Torso Angle



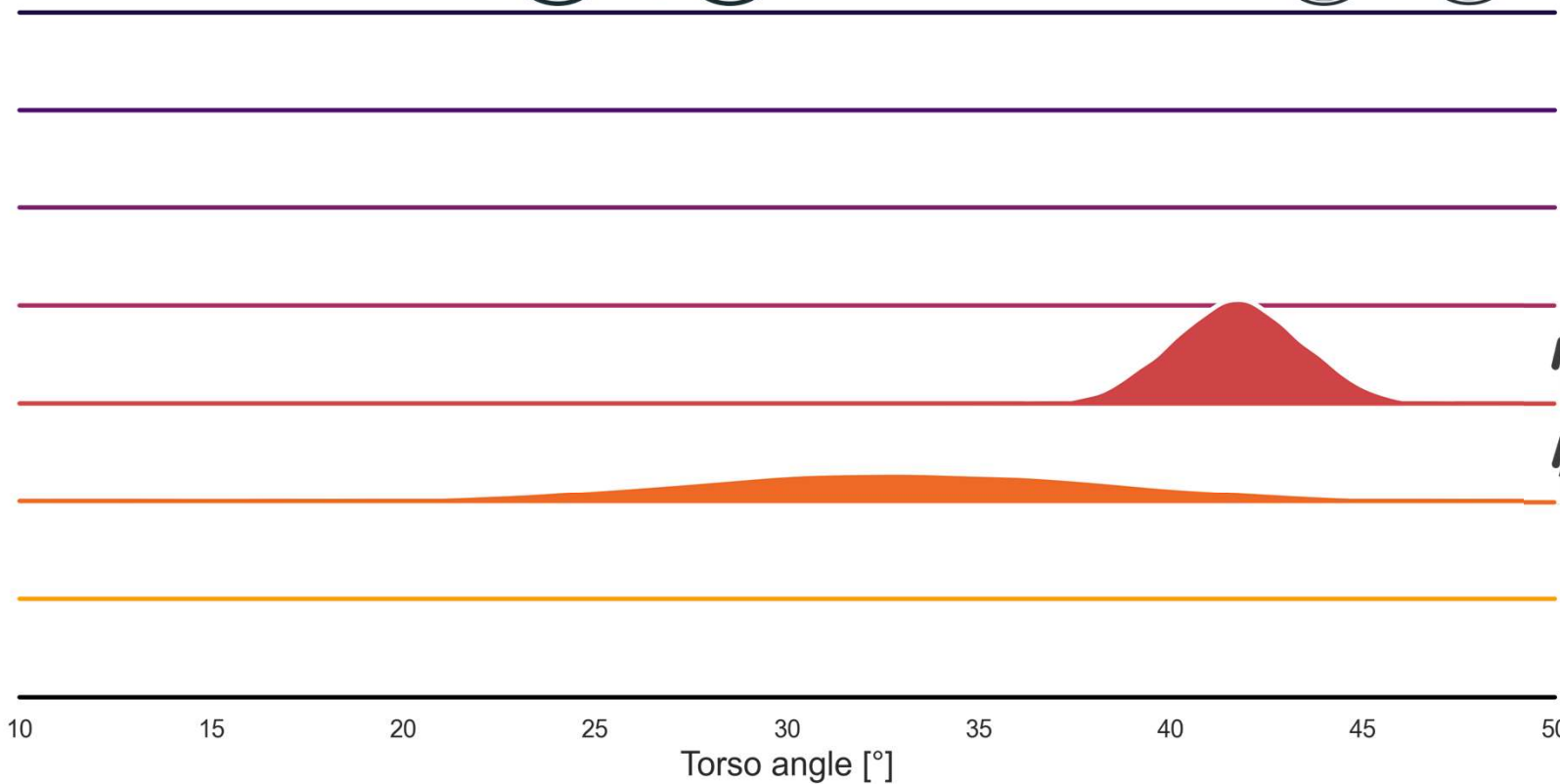
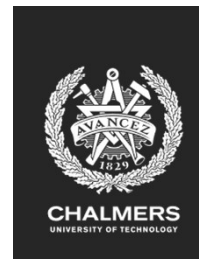
Torso Angle



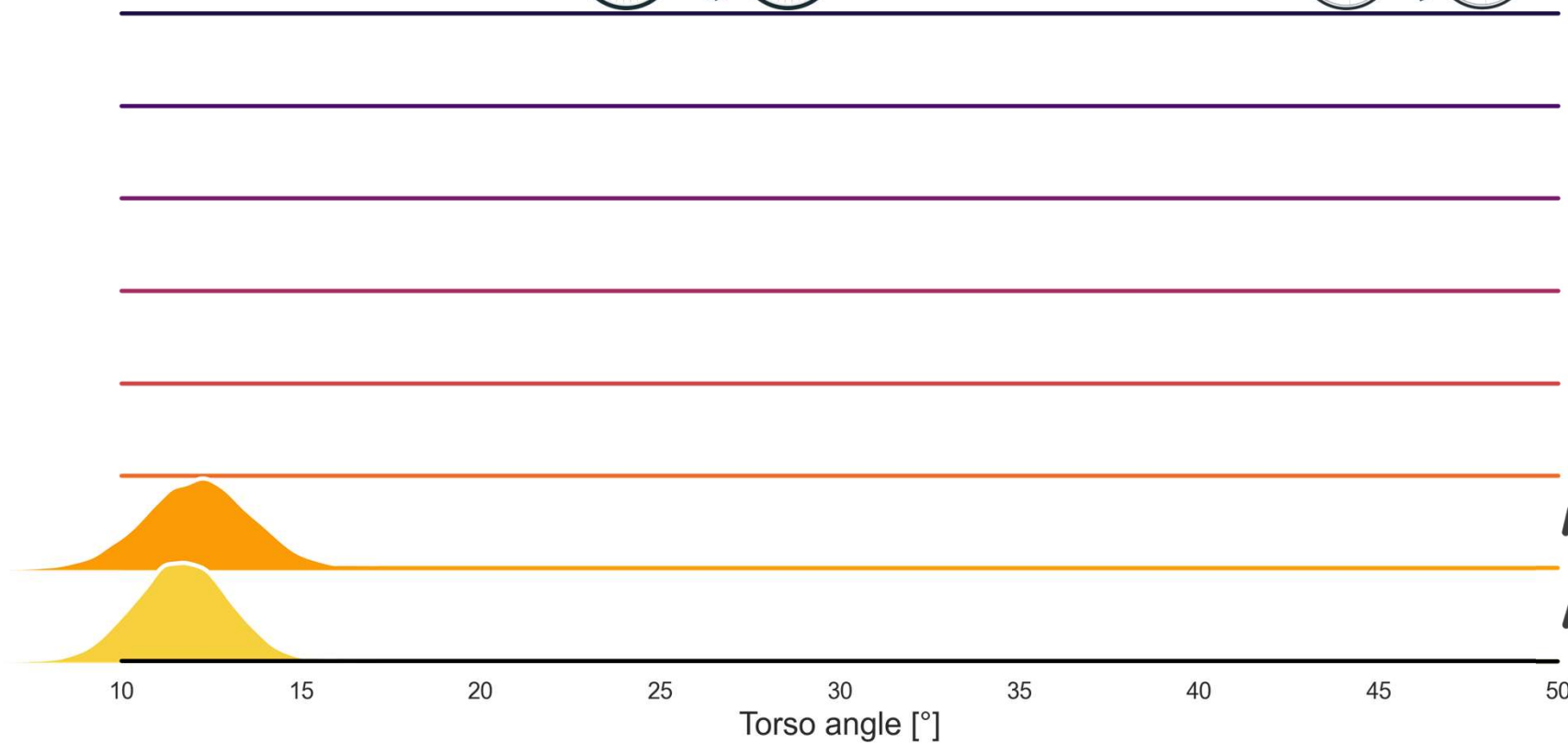
Torso Angle



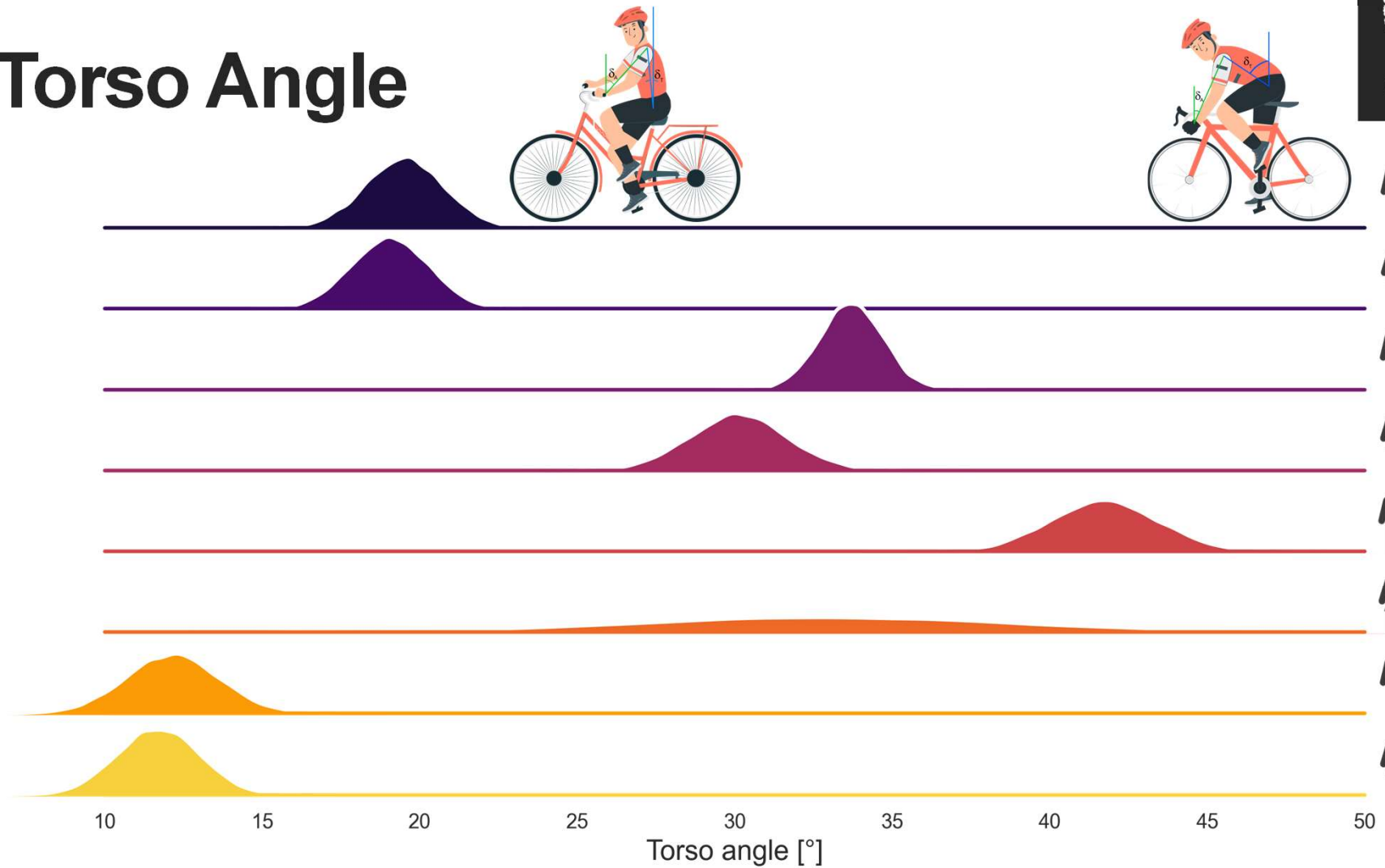
Torso Angle



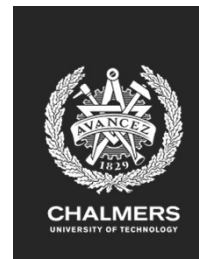
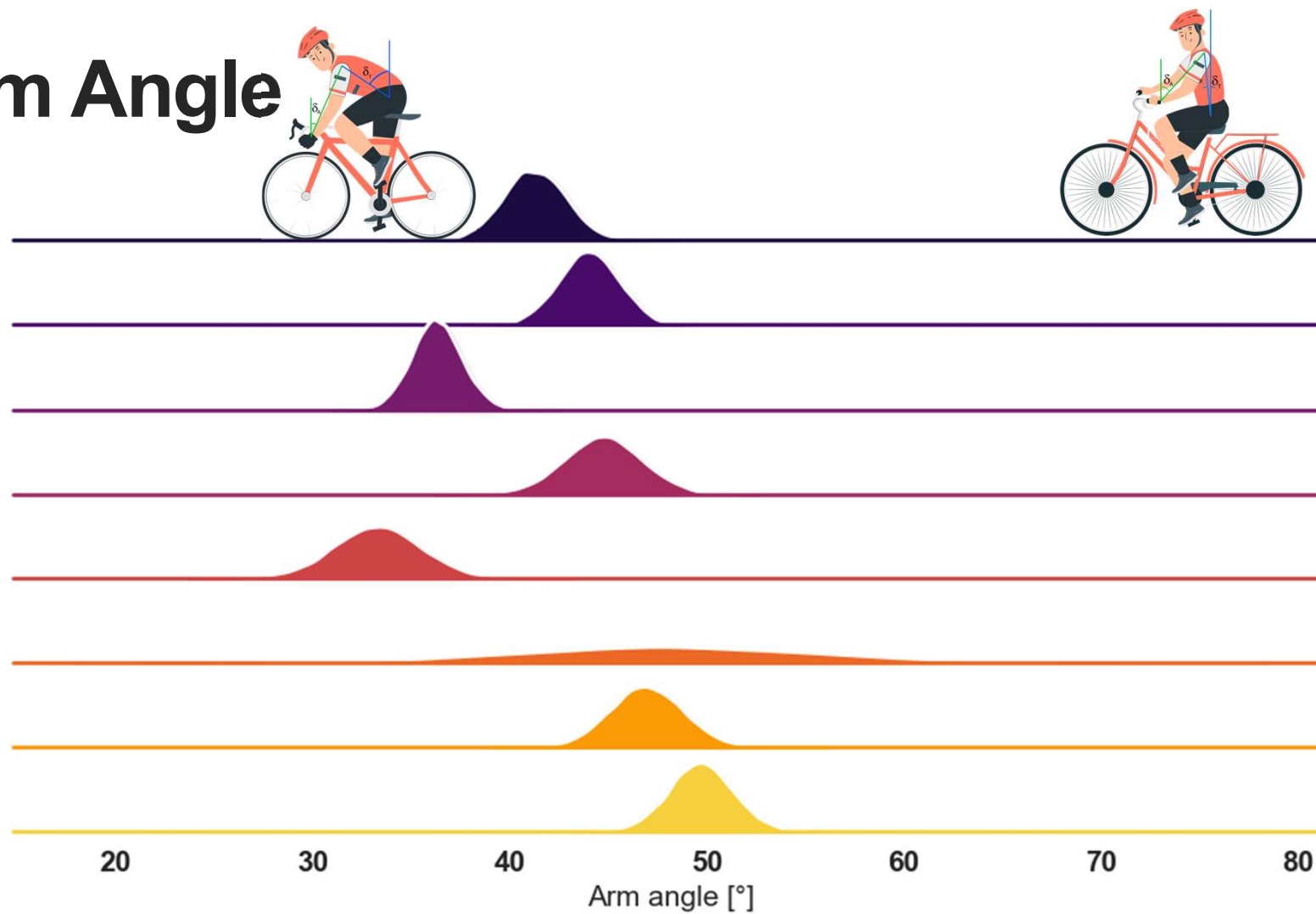
Torso Angle



Torso Angle



Arm Angle



2024-10-28

Summary



150 of 240 were males.



Most common used bikes are classic or hybrid.



30% are e-bikes.



75% wear the helmet.



More females ride a classic or e-bike



Males ride more hybrid/racing bikes.

Summary

- Quantify the angles for each bike type.
- Sex differences are distinct for racing and hybrid configurations.
- Being on an ebike, resulted in the most upright torso position.
- Arm angle varies between 40-50% for all the configurations.





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